CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 7: **Proportional Reasoning-1** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. 72 km/h expressed in m/s is: 1

(a) 24 m/s **(b)** 20 m/s **(c)** 1.2 m/s **(d)** 259.2 m/s

2. Two numbers are in the ratio 6 : 11 and their sum is 68. The larger number is: 1

(a) 34 **(b)** 48 **(c)** 24 **(d)** 44

3. Two numbers are in the ratio 5 : 6. If 2 is added to each, the numbers become 37 and 44. The original smaller number is: 1

(a) 70 **(b)** 35 **(c)** 37 **(d)** 33

4. A mixture of 72 L has milk and water in the ratio 5 : 7. The quantity of water is: 1

(a) 47 L **(b)** 30 L **(c)** 42 L **(d)** 37 L

5. 9 pipes fill a tank in 8 hours. 6 such pipes will fill it in: 1

(a) 12 hours **(b)** 11 hours
(c) 5.333333333333333333333333333 hours **(d)** 13 hours

6. A : B = 2 : 5 and B : C = 10 : 3. If A + B + C = 119, then C equals: 1

(a) 23 **(b)** 21 **(c)** 28 **(d)** 70

7. On a map with scale 1 : 25000, a distance of 8 cm represents: 1

(a) 200 km **(b)** 2 km **(c)** 20 km **(d)** 10 km

8. 90 km/h expressed in m/s is: 1

(a) 25 m/s **(b)** 1.5 m/s **(c)** 30 m/s **(d)** 324 m/s

9. Two numbers are in the ratio 7 : 9 and their sum is 80. The larger number is: 1

(a) 45 **(b)** 35 **(c)** 50 **(d)** 40

10. Two numbers are in the ratio 3 : 8. If 6 is added to each, the numbers become 18 and 38. The original smaller number is: 1

(a) 24 **(b)** 18 **(c)** 12 **(d)** 6

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A map has scale 1 : 100000. Two places are 9 cm apart on the map. Find the actual distance in km. 2

12. Two numbers are in the ratio 3 : 13 and their sum is 64. Find the numbers. 2

13. Convert 126 km/h into m/s. 2

14. A map has scale 1 : 250000. Two places are 5 cm apart on the map. Find the actual distance in km. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A mixture of 66 L contains milk and water in the ratio 5 : 6. How much water must be added to make the ratio 5 : 9? 3

16. A : B = 4 : 5 and B : C = 10 : 3. If A + B + C = 168, find the three numbers. 3

17. The ages of a mother and daughter are in the ratio 6 : 1. The mother is 42 years old. Find their ages 7 years ago and the ratio then. 3

OR

A mixture of 80 L contains milk and water in the ratio 3 : 1. How much water must be added to make the ratio 3 : 2? 3

18. A : B = 2 : 5 and B : C = 10 : 9. If A + B + C = 190, find the three numbers. 3

OR

The ages of a mother and daughter are in the ratio 4 : 1. The mother is 36 years old. Find their ages 3 years ago and the ratio then. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. 16 men can complete a piece of work in 21 days. 5
(i) How many days will 12 men take?
(ii) How many men are needed to finish it in 14 days?

OR

Anil and Bhanu start a business with ₹ 16000 and ₹ 24000. They share a profit of ₹ 1200 in the ratio of their investments. Find the ratio of investments and each person's share. 5

20. 25 men can complete a piece of work in 12 days. 5
(i) How many days will 20 men take?
(ii) How many men are needed to finish it in 15 days?

OR

Anil and Bhanu start a business with ₹ 9000 and ₹ 15000. They share a profit of ₹ 960 in the ratio of their investments. Find the ratio of investments and each person's share. 5

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND